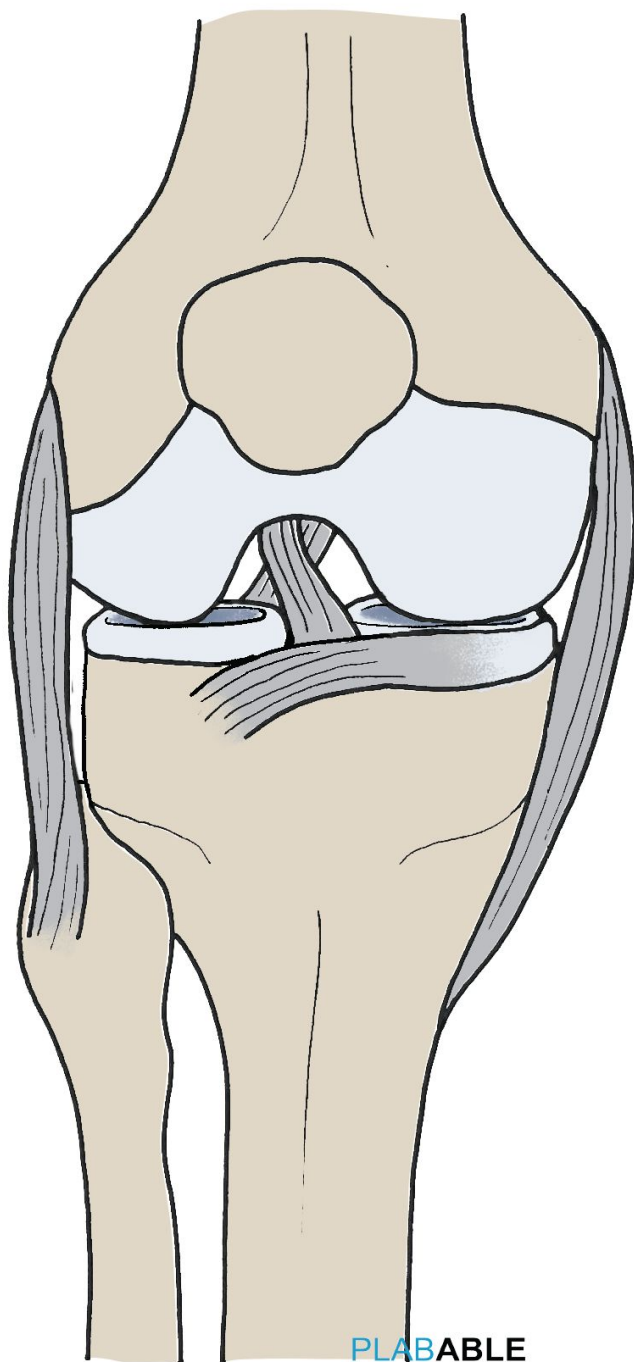


PLABABLE

GEMS

VERSION 1.0

ORTHOPAEDICS



PLABABLE

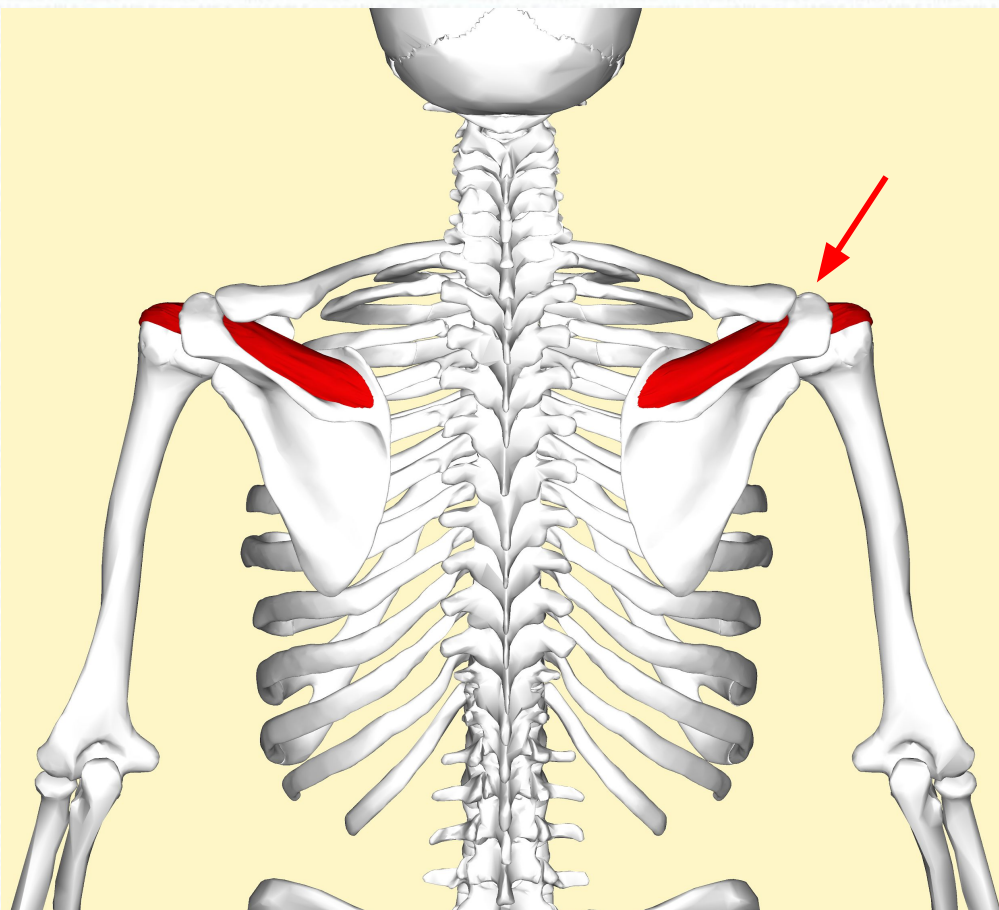
Shoulder Impingement Syndrome

Presentation

- Pain and weakness of the shoulder movements especially during overhead activities
- Restriction of activity
- Most common cause: narrowing of the Subacromial space causing **supraspinatus tendinitis**
- H/O repetitive overhead activity or carrying heavy weights

Management

- Conservatively: NSAIDs, physiotherapy



Frozen Shoulder

Presentation

- Pain and stiffness of the shoulder
- Restriction of both active and passive shoulder movements
- Inability to do passive external rotation (specific)
- Caused by thickening and contraction of the glenohumeral joint capsule and formation of adhesions (**adhesive capsulitis**)



Management

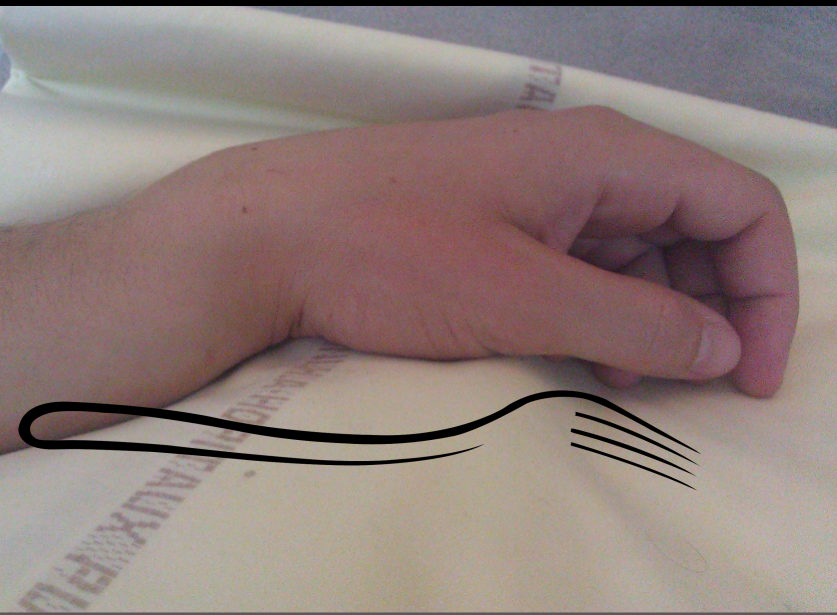
- Paracetamol and NSAIDs
- Physiotherapy with joint mobilisation and stretching
- Steroid injection

Colles Fracture

Presentation

- Radial fracture with or without ulnar fracture
- **Dinner-fork** deformity (distal fragment deviated backwards and laterally)
- Fall on an outstretched hand
- Common in elderly

Dinner-fork Deformity

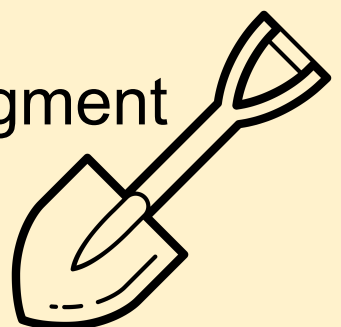


Treatment

Closed reduction and immobilisation with Colles cast

Smith's Fracture

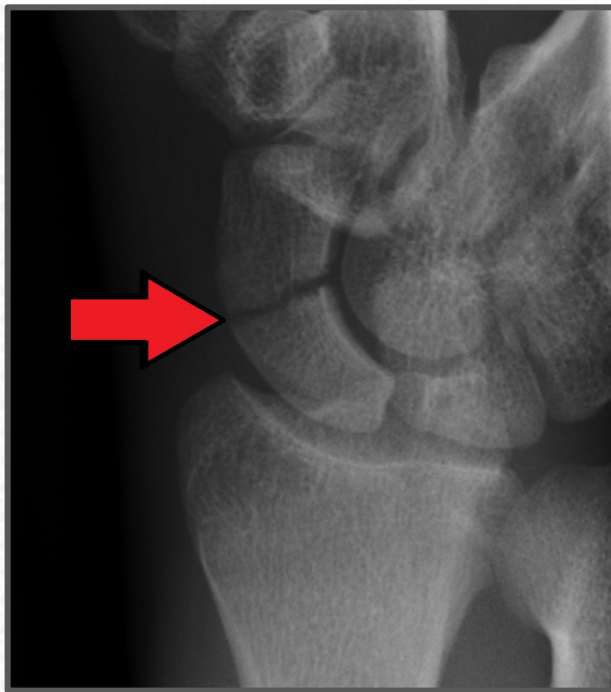
- Reverse colles fracture
- Anterior displacement of the distal fragment
- **Garden spade** deformity



Scaphoid Fracture

Presentation

- Tenderness in the anatomical snuffbox
- Fall on an outstretched hand
- High risk for **avascular necrosis** as blood supply is from distal to proximal part
- Most common carpal bone to fracture



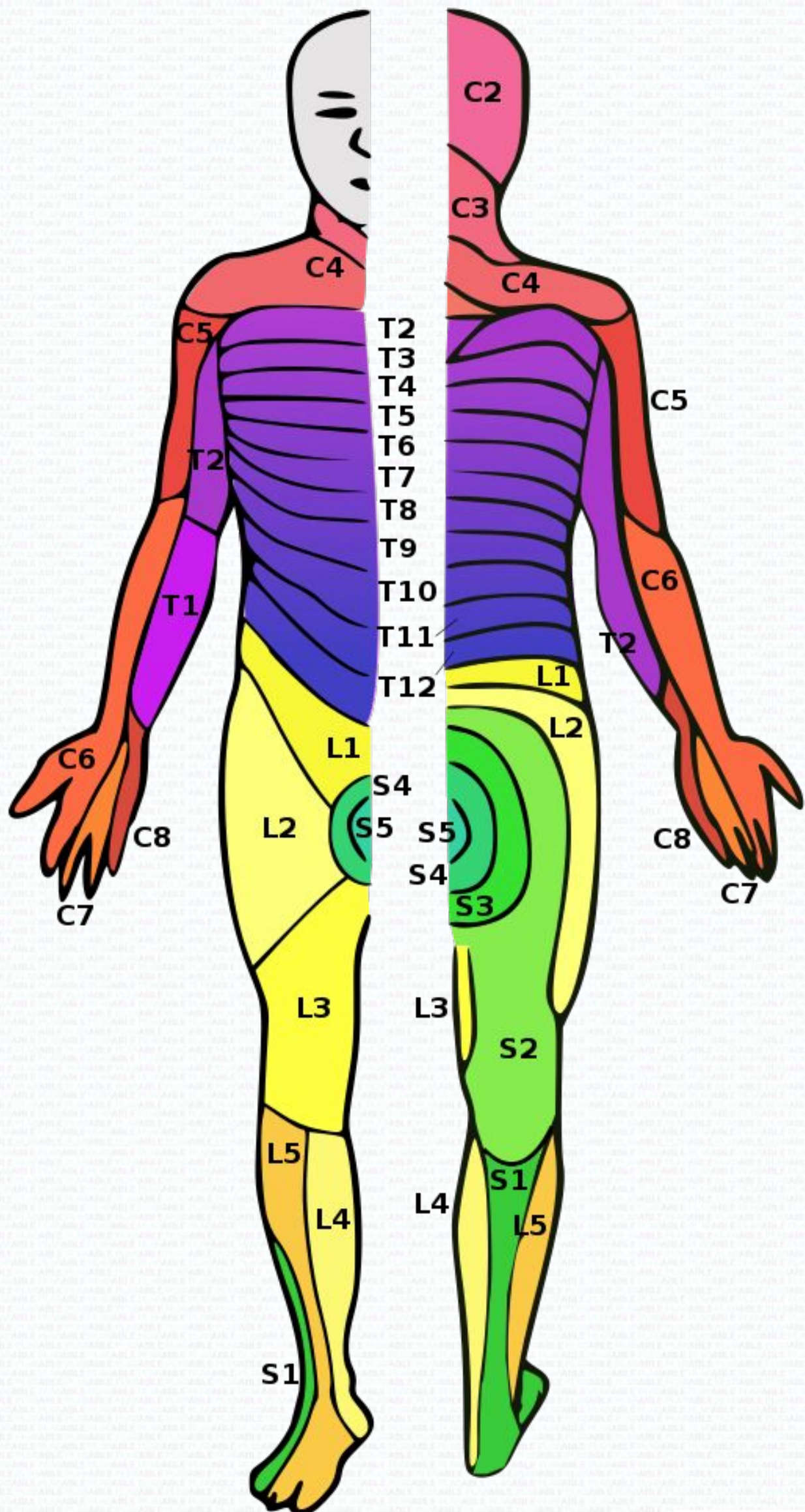
Investigations

- **X-ray:** One immediately and one 10-14 days later if the first one does not show the fracture line

Treatment

- **# visible on X-ray:** Scaphoid cast for 6 weeks
- **# not visible on X-ray:** Cast immobilisation of the wrist and review after 2 weeks with X-ray

Dermatomes



Complication of Fracture

Fat embolism syndrome

- Seen with fracture of long bones
- Breathlessness and chest pain
- Petechial rash
- Oliguria
- Haematuria

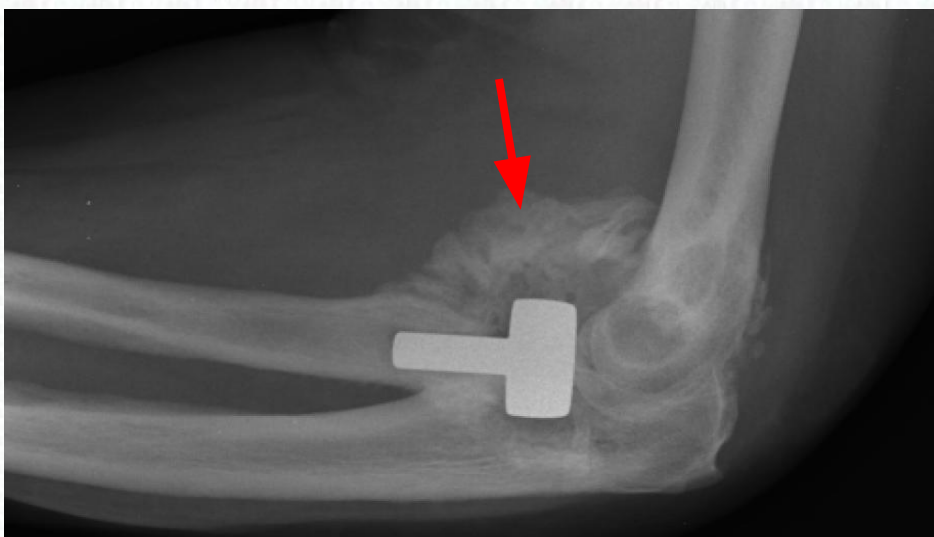


Compartment syndrome

- Increased compartment pressure
- Decreased or no blood flow
- Pain and pulseness

Gunstock deformity: Cubitus varus following supracondylar fracture of the humerus

Myositis Ossificans: Formation of new bone in the muscle following fracture



Osteosarcoma

Features

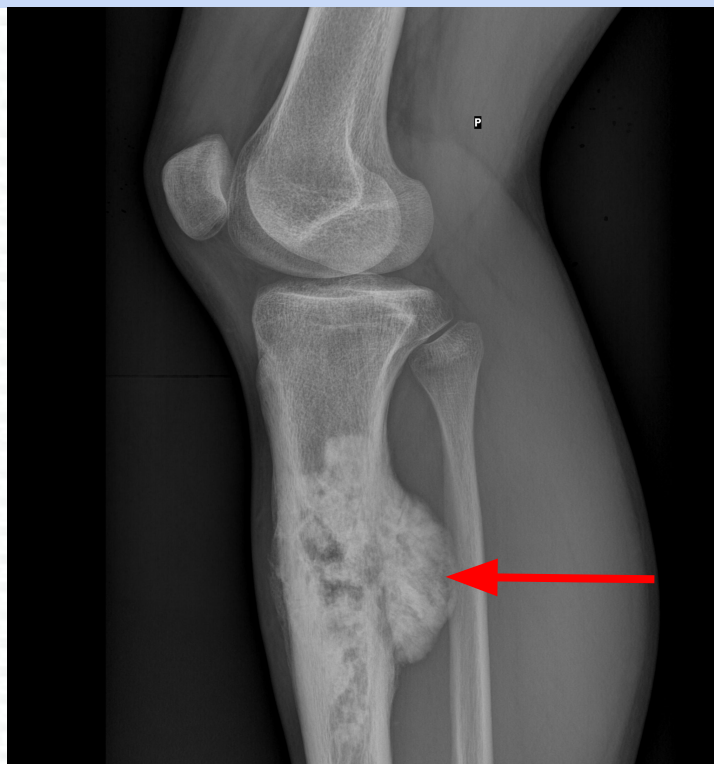
- Knee pain increasing at night
- Visible swelling in large tumour
- Fracture even with minor trauma
- **Commonly affect:** distal femur and proximal tibia
- **Bimodal distribution:** adolescence & old adults

Investigation

- **X-ray:**
 - **Sunburst appearance**
 - Codman's Triangle
- CT scan and bone biopsy

Treatment

- Radical resection of the tumour



Osteoid Osteoma

Brain trainer:

A young man with unilateral leg pain which is worse at night. The pain is relieved by NSAID. What is the most likely diagnosis?

→ Osteoid osteoma

Supraspinatus Tendinitis

Brain trainer:

A male complains of right arm shoulder pain which is both worse at night and when performing activities which involve lifting above the shoulder. He is a tennis player. There is no history of trauma. What is the most likely diagnosis?

→ **Supraspinatus tendinitis**

Achilles Tendon Rupture

Brain trainer:

A middle aged man playing football feels a pop at the back of his left leg followed by pain at his calf. He has not played sport for sometime. What is the most likely diagnosis?

➔ **Achilles tendon rupture**

Transient Synovitis

Brain trainer:

A 6 year old child presents with acute onset hip pain, a limp and a mild fever. He is otherwise well. What is the most likely diagnosis?

→ Transient synovitis

Elbow Fracture

Brain trainer:

After falling on an outstretched arm, a person presents with pain, swelling and limited range of movement of the elbow. Which part of the radius is most likely fractured?

→ If child → radial neck

→ If adult → radial head

Fat Embolism

Brain trainer:

After a fracture of the femur a patient presents with a mild fever, heart rate 140/min, decreased level of consciousness and hypoxaemia. What is the most likely diagnosis?

➔ **Fat embolism**

Bone Metastasis

Brain trainer:

What is the most common origin of a bone metastasis?

→ Male → prostate > lung

→ Female → breast > lung

Elbow Fracture

Brain trainer:

What is the most common elbow fracture in children?

→ **Supracondylar fractures of the humerus**

Torus Fracture

Brain trainer:

What is the most common type of fracture and the most common fracture site in childhood?

→ **Buckling (torus) fracture of the distal radius**

Ewing Sarcoma

Features

- Bone pain
- Fever
- Weight loss
- Fractures
- Common in the **second decade** of life
- **Most common location:** diaphysis of femur, tibia and humerus

Management

- Chemotherapy plus surgery or radiotherapy as it is highly metastatic

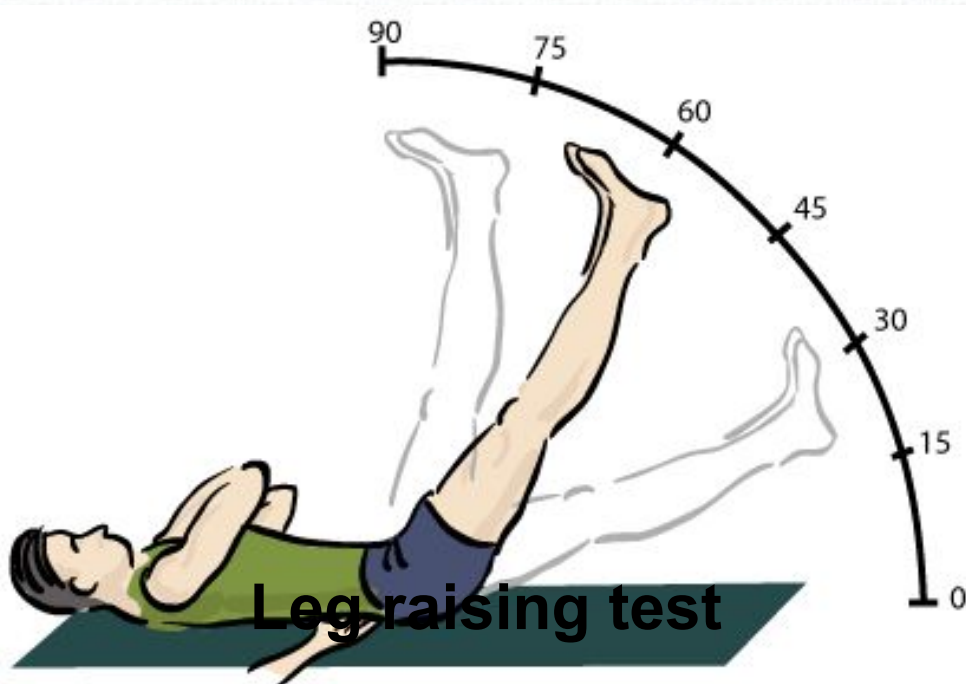
Sciatica

Presentation

- **Shooting pain** spreading from buttock to the leg
- Numbness, paraesthesia, weakness and/or loss of tendon reflexes
- Common cause: **disk prolapse** (L4, L5 or S1)
- Pain ↓ on lying down and ↑ on prolonged sitting or walking
- Positive leg raising test
- Large herniation can cause cauda equina syndrome

Management

- NSAIDs
- Amitriptyline or gabapentin (neuropathic pain)
- Cauda equina syndrome (needs urgent surgical decompression)



Knee Injuries

Unhappy triad is injury to:

- Anterior cruciate ligament
- Medial collateral ligament
- Medial meniscus

Most commonly seen in contact sport (sudden twisting of the players leg)

Investigations

- MRI (gold standard)

Management (PRICER):

- **P** - Protect
- **R** - Rest
- **I** - Ice
- **C** - Compression
- **E** - Elevation
- **R** - Rehabilitation

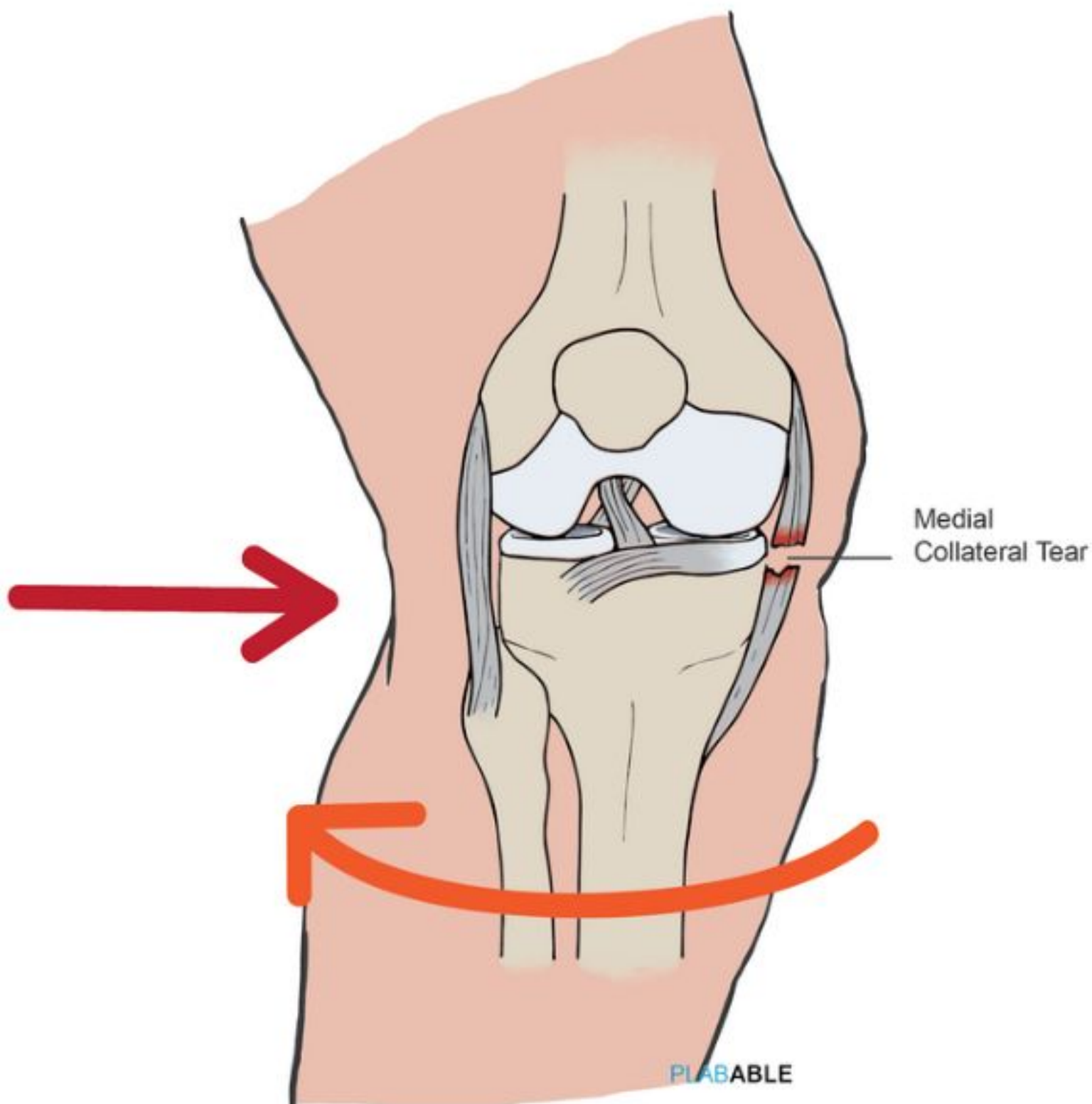
Arthroplasty and repair in case of complete ligament tear



Knee Injury Mechanism

Medial collateral ligament

- Direct blow to the lateral side of the knee or a twisting injury. Often occurs along with meniscal tears

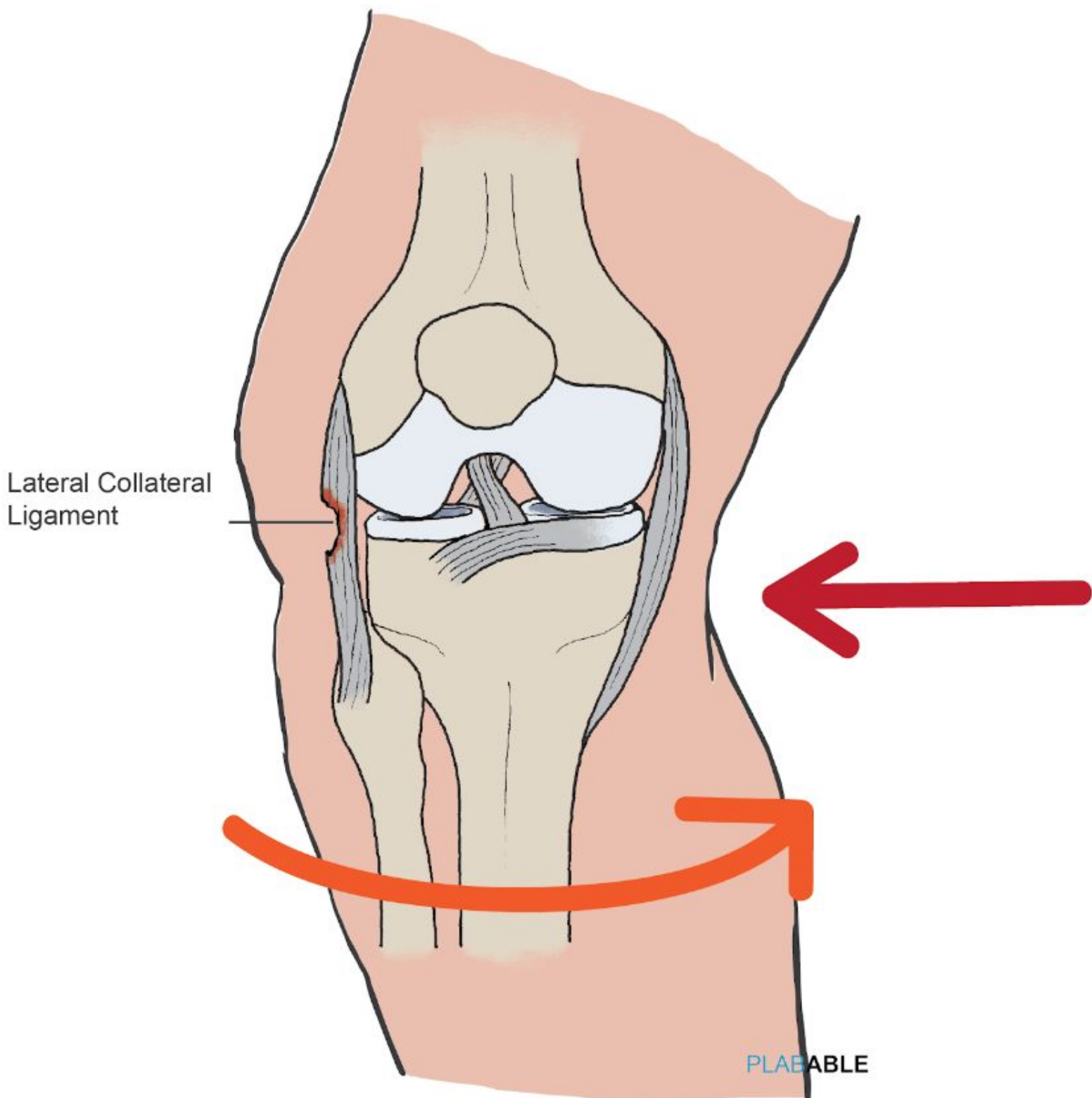


Medial Collateral Ligament Injury

Knee Injury Mechanism

Lateral collateral ligament

- Direct blow to medial side of the knee or a runner twisting to the direction of the planted foot (twisting inwards)

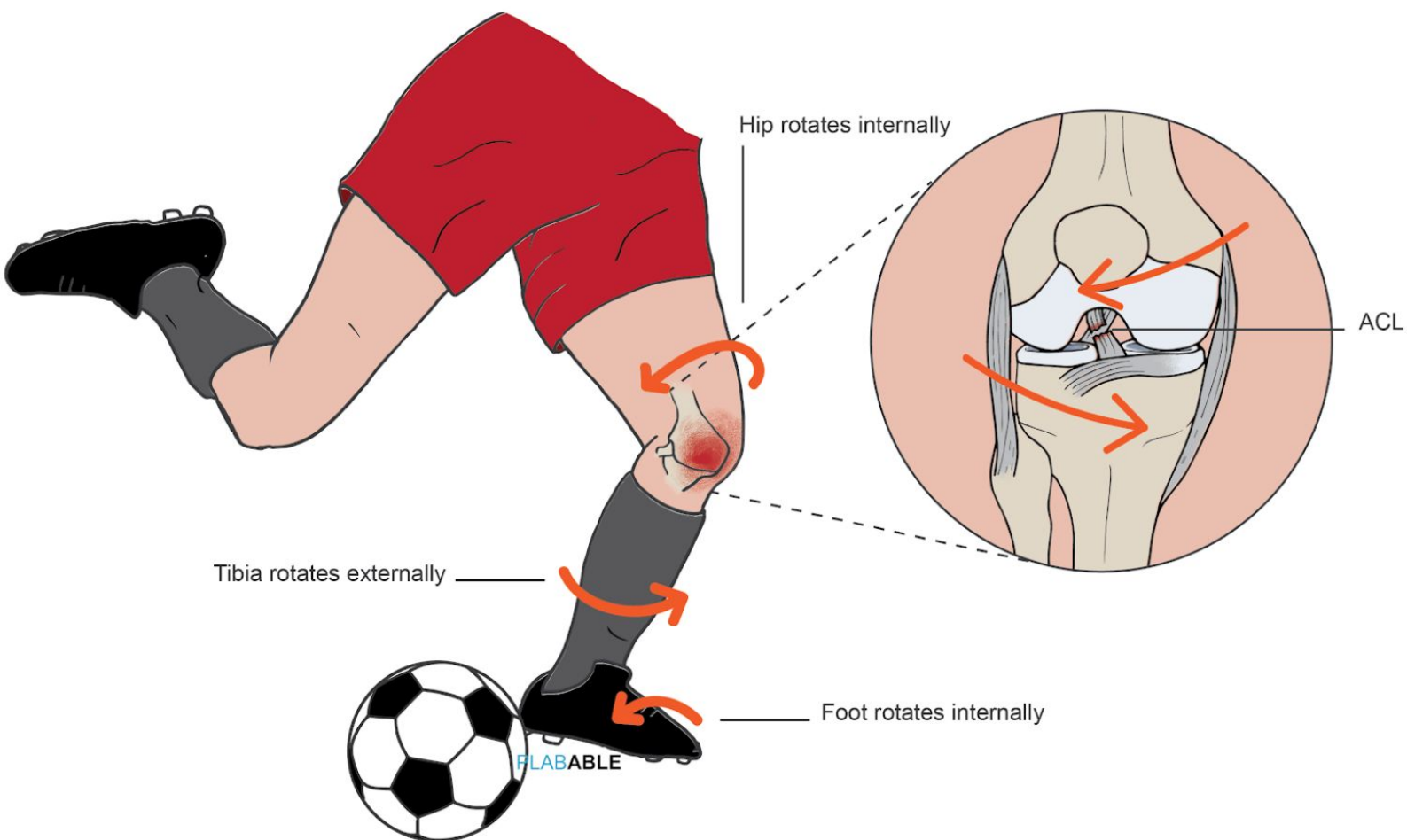


Lateral Collateral Ligament Injury

Knee Injury Mechanism

Anterior cruciate ligament

- Deceleration injury or when the athlete lands hard on the leg and quickly pivots to the opposite direction
- ACL tears usually have a stem where the foot is fixed on the ground when a rotational force is applied which is followed by a “pop” sound. They would complain that the knee “gives way”



Knee Injury Mechanism

Posterior cruciate ligament:

- Hyperflexion injury or a fall onto a flexed knee or if the knee hits the dashboard of a car during a road traffic accident
- Look for a history of direct impact on the shin (specifically the proximal tibia) when the knee is bent



Meniscal Tear

Brain trainer:

What associations with meniscal tears do you need to know for the exam ?

- ➔ **Twisting or pivoting, a popping sensation heard at time of injury**
- ➔ **Often associated with ACL injury**

Treatment of Sprain Injury

Brain trainer:

With a sprain injury, what does pay the “p.r.i.c.e” mean?

→ Protect

→ Rest

→ Ice

→ Compress

→ Elevate

Nerve Injuries

Traction injury at birth	Brachial plexus
Fracture of humeral neck Shoulder dislocation	Axillary nerve
Fracture of humeral shaft (Spiral groove)	Radial nerve
Elbow dislocation	Ulnar nerve or Median nerve
Monteggia fracture	Radial nerve (Wrist drop)
Fibular neck fracture	Common peroneal nerve (Foot drop)
Femur neck or acetabular fracture Hip dislocation	Sciatic nerve

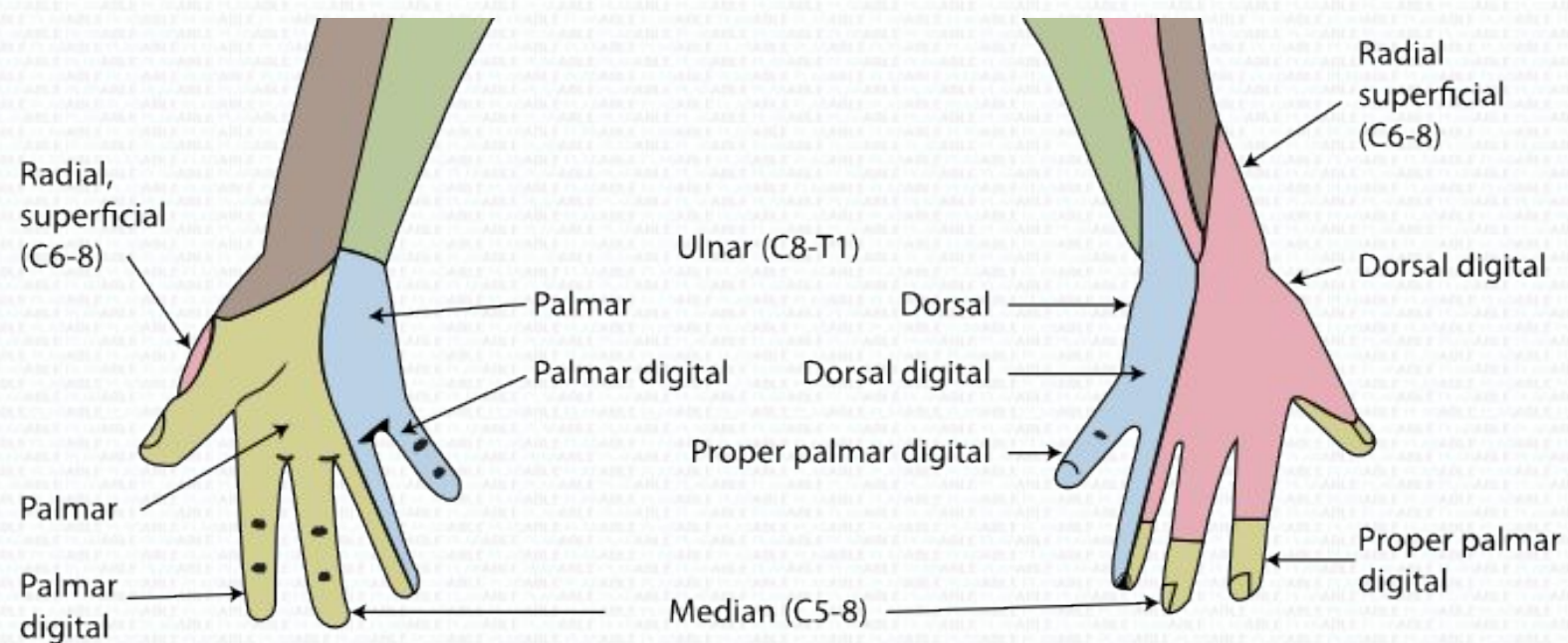
Radial Nerve Injury

Fracture of the humeral shaft: (radial groove)

- Wrist drop
- Triceps is spared
- Loss of sensation to posterior arm and forearm
- Sensory loss to postero-medial 3 and a half fingers below the fingertips

Lesion at wrist

- Loss of finger extension
- Normal wrist movement
- No loss of sensation



Ulnar Nerve Injury

High ulnar palsy (cubital tunnel syndrome at elbow):

- Weakness in flexion of 4th and 5th fingers
- Paralysis of interossei and medial two lumbricals
- **Ulnar claw hand**
- Atrophy of hypothenar muscles
- Sensory loss in the medial one and a half finger anterior and posterior

Low ulnar palsy (wrist):

- Pins and needles in the medial one and a half fingers
- Loss of motor function later
- Seen in cyclists
- Also, known as Guyon's canal syndrome or ulnar tunnel syndrome

Carpal Tunnel Syndrome

Presentation

- Tingling, numbness and pain in the distribution of the median nerve
- Weakness in hand grip and opposition of the thumb

Causes

- Pregnancy
- Wrist trauma
- Obesity
- Hypothyroidism
- Renal failure

Tests

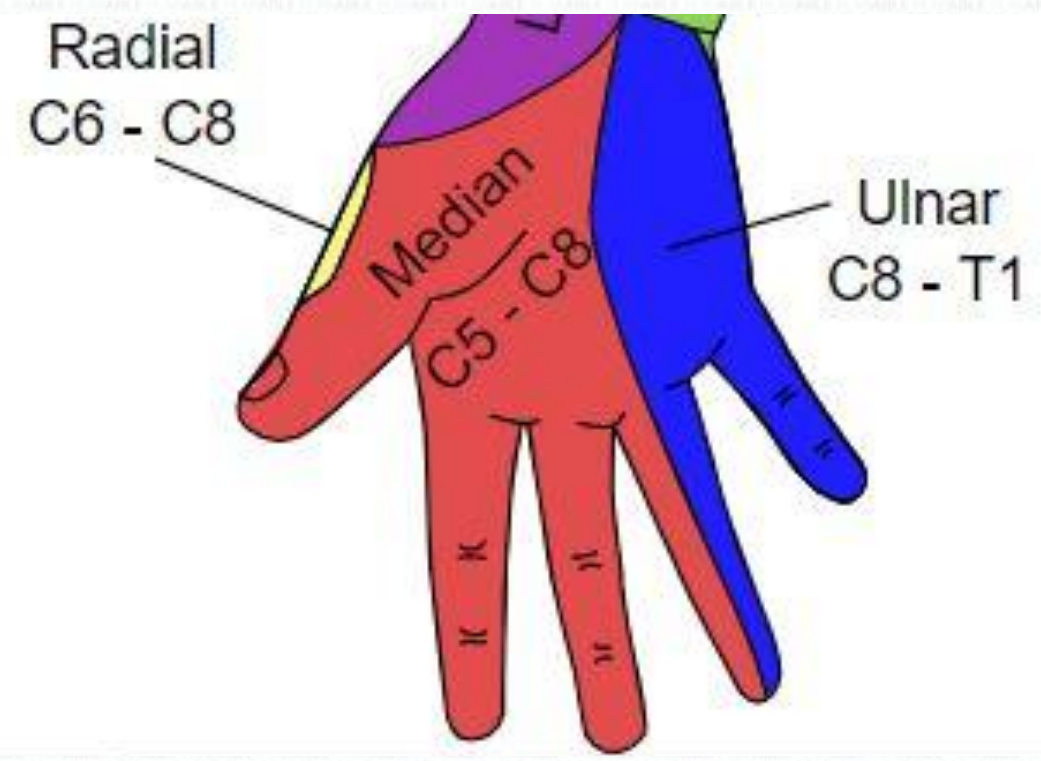
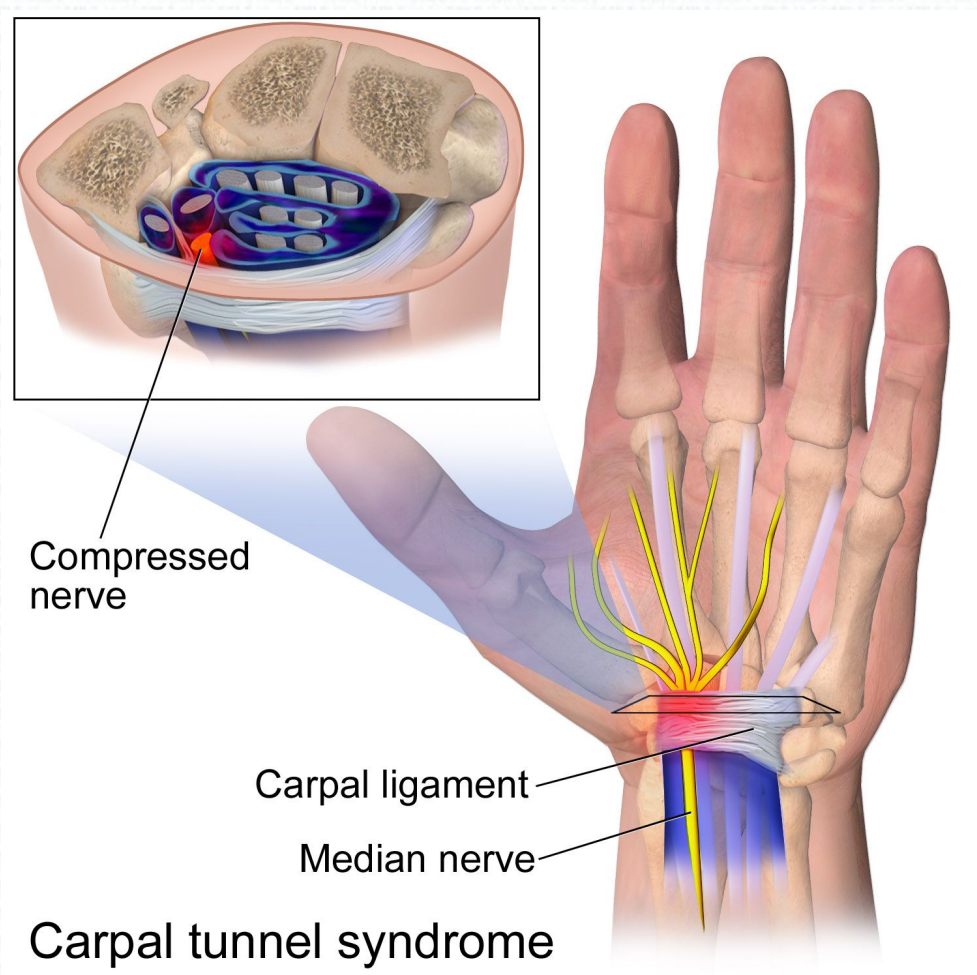
- Positive Tinel's sign
- Positive Carpal Tunnel compression test
- Electroneurography (gold-standard)

Management

- Activity modification and local steroid injection
- Release of flexor retinaculum

Carpal Tunnel Syndrome

Compression of the median nerve in the carpal tunnel



Developmental Hip Dysplasia

Presentation

- Asymmetrical gluteal or thigh skin folds
- Limb length discrepancies
- Limitation of hip abduction
- More common in female infants
- Breech presentation ↑ risk

Tests

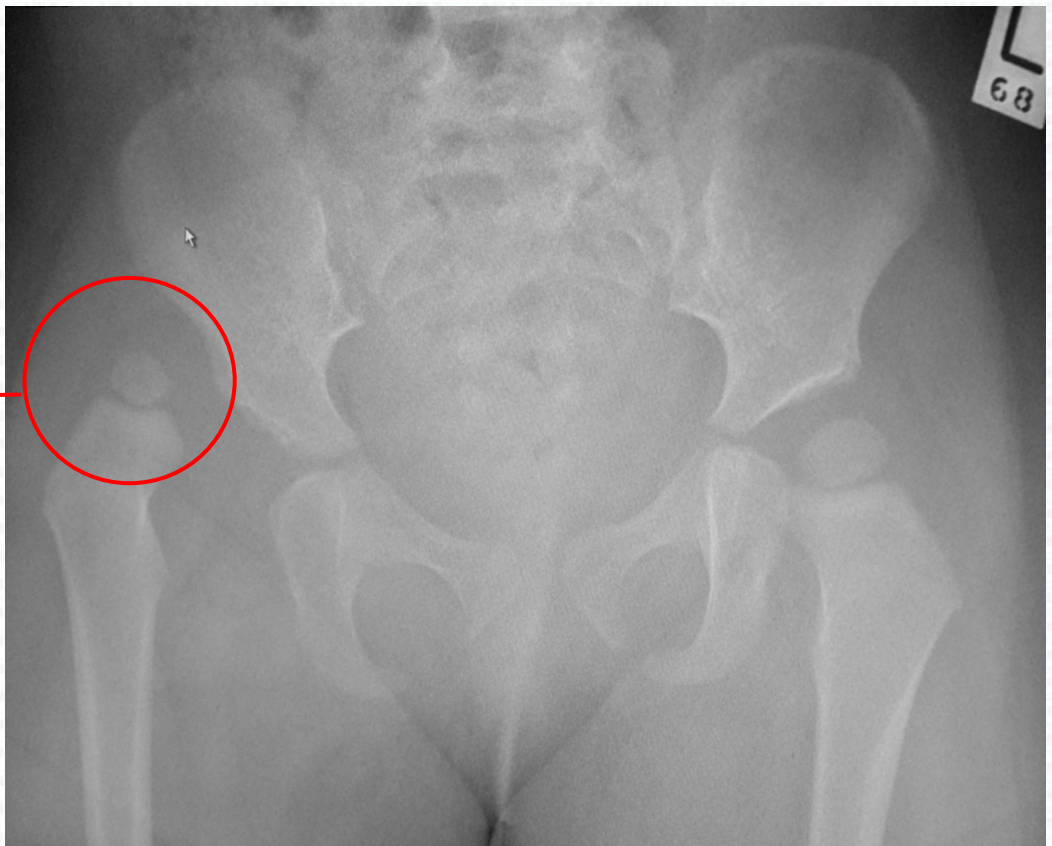
- **Barlow test:** attempts to dislocate the femoral head
- **Ortolani test:** attempts to relocate a dislocated femoral head
- Ultrasound scan of the hips

Management

- Pelvic harness (If age <6 months)
- Surgery (If age >6 months and failed bracing)

Developmental Hip Dysplasia

Formerly known as **congenital dislocation of the hip** is a dysplasia of the hip causing neonatal hip instability



Dislocated
hip in a
child with
dysplasia of
the head

Legg-Calve-Perthes' Disease

Avascular necrosis of the femoral head

Presentation

- 4 to 8 yrs old boy (common)
- Limping
- Pain in the hip or knee
- Limited internal rotation

Investigation

- X-ray - joint space widening
- MRI hip

Management

- Surgery
- Physiotherapy



Paget's Disease

Excessive bone breakdown followed by disorganised new bone formation

Presentation

- Bone pain
- Pathological fracture
- Deafness and tinnitus (compression of CN VIII)

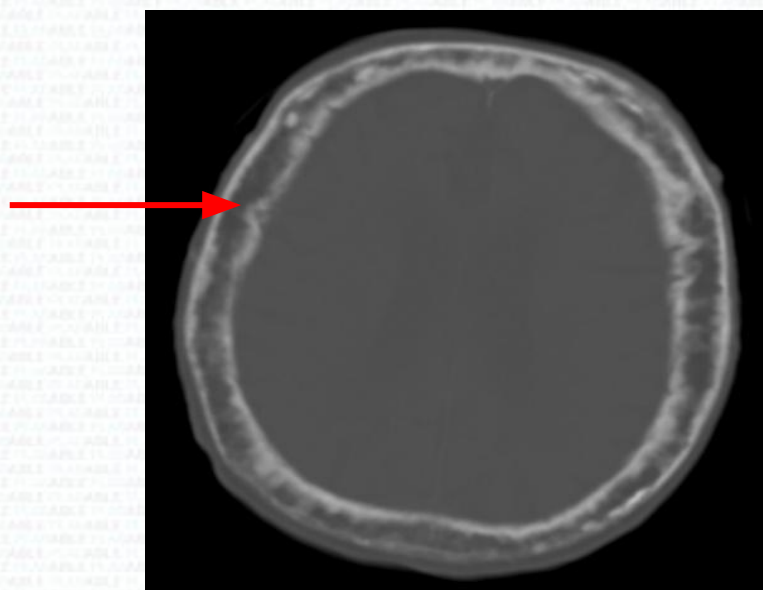
Labs

- ↑ Alkaline phosphatase
- Normal calcium, phosphate and PTH
- **X-ray:**
 - Cotton wool pattern in the skull
 - Osteolytic lesions

Management

- Paracetamol and NSAIDs for pain
- Bisphosphonates to reduce bone resorption - zoledronic acid

Cotton wool pattern in the skull



Fracture Neck of Femur

Presentation

- H/O fall in an elderly or osteoporotic bone
- Pain in the upper thigh or groin
- Inability to bear weight
- Adducted and externally rotated leg
- High risk for avascular necrosis of the femoral head

Investigations

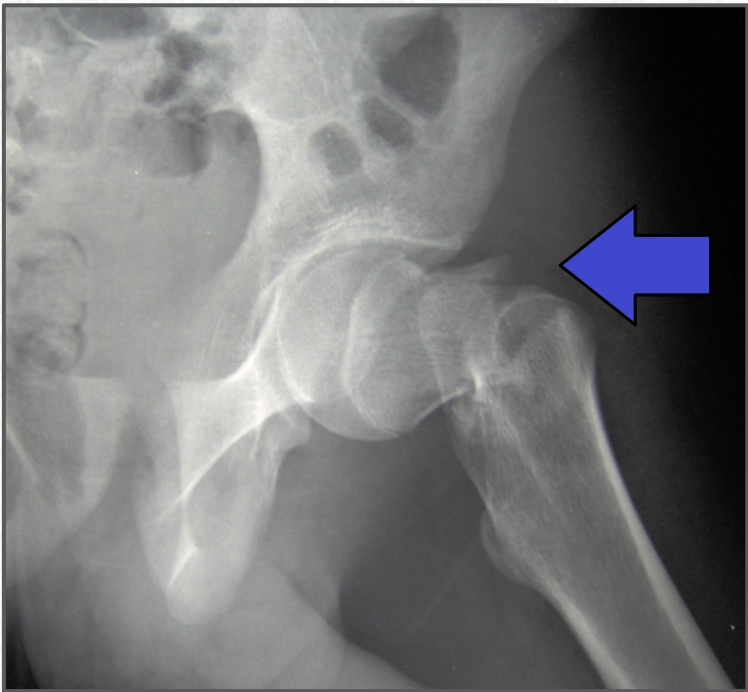
- X-ray - Broken Shenton's line
- MRI if X-ray is normal and fracture is suspected

Management

- Internal fixation of the femoral head
- Total Hip Replacement in case of AVN

Fracture Neck of Femur

Femoral neck #



Total hip replacement

Osteoarthritis

Presentation

- Joint pain ↑ during activity and ↓ by rest
- Commonly affects weight bearing joints such as Hip and Knee
- Joint crepitus and tenderness
- **Bone swelling:**
 - Heberden's nodes (DIP)
 - Bouchard's nodes (PIP)

Risk factors: Old age and overweight

X-ray

- **L** - Loss of joint space
- **O** - Osteophytes
- **S** - Subchondral cysts
- **S** - Subchondral sclerosis

Management

- Pain management: paracetamol, NSAIDs
- Weight reduction and physiotherapy
- Joint replacement surgery (last resort)

Mallet Finger

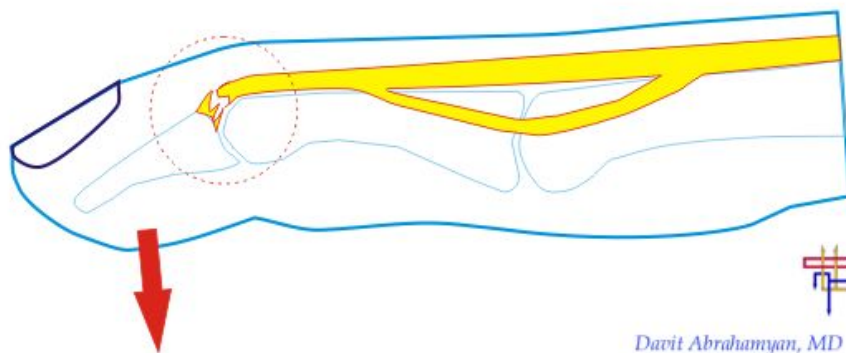
Rupture of extensor tendon of the finger

Presentation

- Fixed flexion deformity at DIP joint
- Swelling and ecchymosis
- H/O trauma

Management

- Partial tear: splint
- Complete tear: surgical fixation



Jersey finger: Rupture of Flexor digitorum profundus causing loss of flexion

Trigger Finger

- Also known as stenosing flexor tenosynovitis
- The finger becomes 'locked' after flexion
- Difficulty in straightening the finger without pulling on it by the other hand.

Management

- Splinting
- Steroid injection
- Surgery - widening of the tendon sheath

Gamekeeper's or Skier's thumb

- Rupture of the Ulnar collateral ligament
- Thumb is hyperextended and deviated laterally

Osteoporosis

Bone mineral density ≤ -2.5 from the mean

Risk factors

- Increasing age
- Female sex (post-menopausal)
- Corticosteroid use
- Cushing's syndrome
- Primary hyperparathyroidism

Presentation

- Fracture with even low trauma



Investigations

- DEXA scan
- Normal calcium and PTH

Management

- Bisphosphonates - alendronate (first-line)
- Calcium and Vit D supplements
- Denosumab
- Strontium ranelate

Tennis Elbow

Lateral epicondylitis at the common origin of extensor forearm muscles due to repeated activity

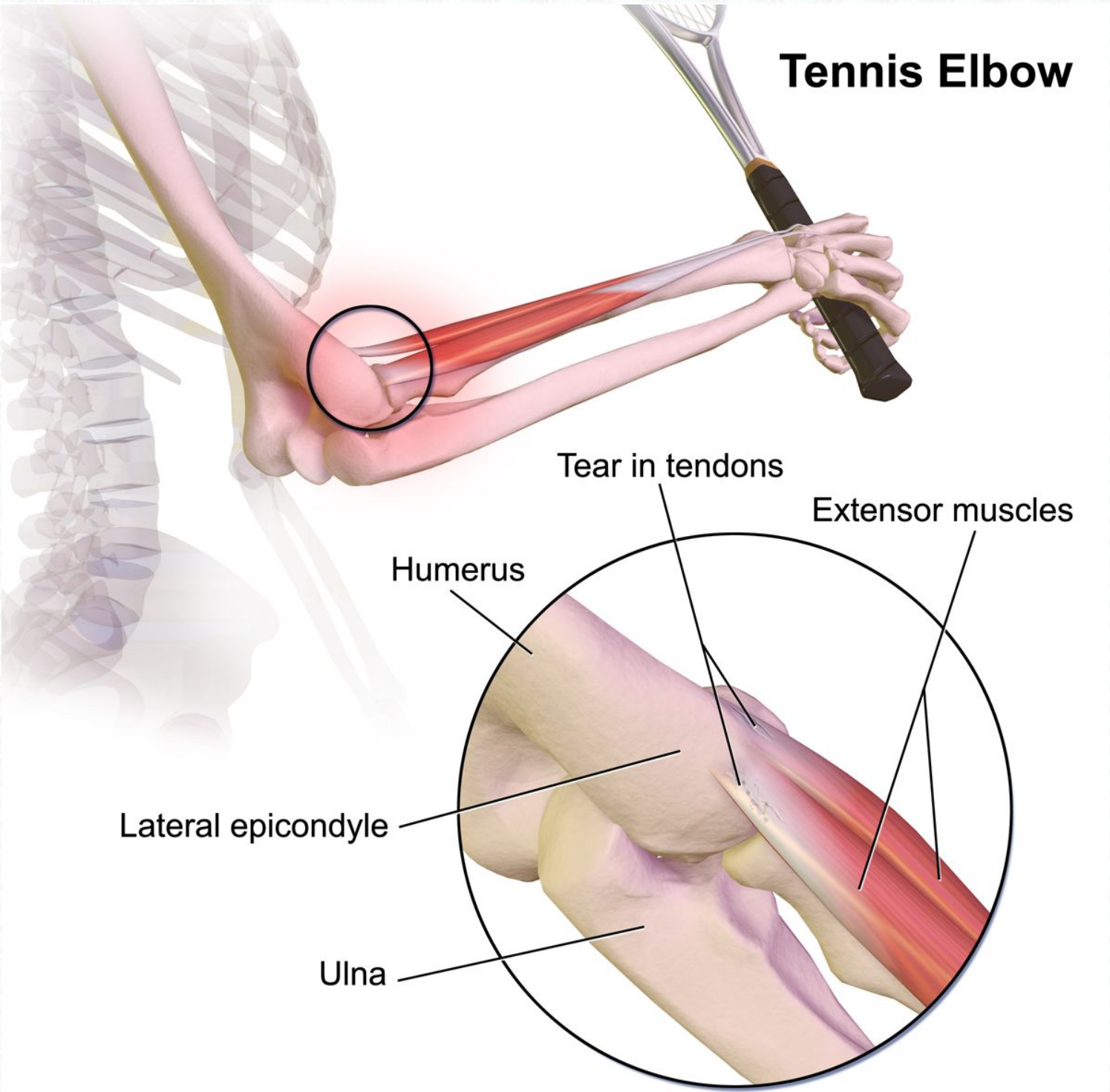
Presentation

- Pain and tenderness at the **lateral epicondyle** of the humerus
- Positive Cozen's test (pain on resisted extension of the wrist)

Management

- Activity modification
- NSAIDs
- Local steroid injection
- Physiotherapy
- Surgical release of the extensors (last resort)

Tennis Elbow



Golfer's Elbow

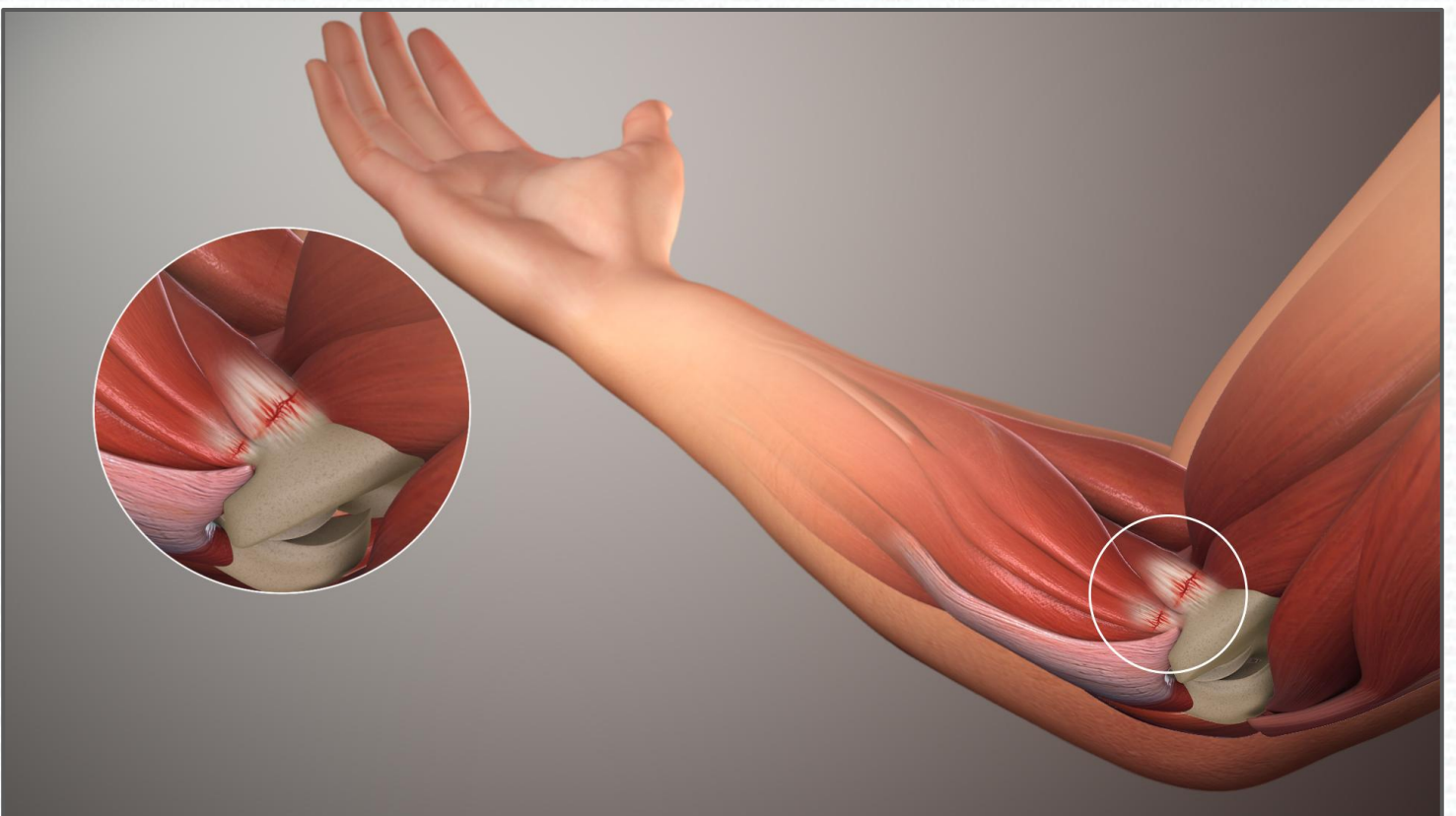
Medial epicondylitis at the common origin of flexor forearm muscles due to repeated activity

Presentation

- Pain and tenderness at the medial epicondyle of the humerus
- Pain increases on wrist flexion and pronation

Management

- Activity modification
- NSAIDs
- Local steroid injection
- Physiotherapy
- Surgical release of the flexors (last resort)



Septic Arthritis

Presentation

- Single swollen and red joint
- Pain on active or passive movement
- Fever and rigors
- Common organisms:
 - Overall - *S.aureus*
 - Sexually active adults - *N. gonorrhoea*
- Commonly involved joint is knee

Investigation

- Aspiration of synovial fluid
- Blood culture

Management

- Flucloxacillin for 4-6 weeks
- Cefotaxime or ceftriaxone (gonococcal arthritis)

Baker's Or Popliteal Cyst

Presentation

- Popliteal mass and pain
- Transillumination +
- Cyst rupture resulting in calf swelling and pain

Management

- Diagnosis is confirmed by USG
- Most cases are treated conservatively
- Aspiration of fluid
- Surgical removal of the cyst (last resort)

Slipped Capital Femoral Epiphysis

Fracture through the growth plate causing slippage of the overlying end of the femoral metaphysis

Presentation

- Pain or discomfort in the groin or knee
- Common in obese adolescent boys
- Limp on the affected side

Investigation

- X-ray in frog leg position shows the widening of the epiphyseal line or displacement of the femoral head

Management:

- Surgical fixation through the growth plate

Slipped Capital Femoral Epiphysis

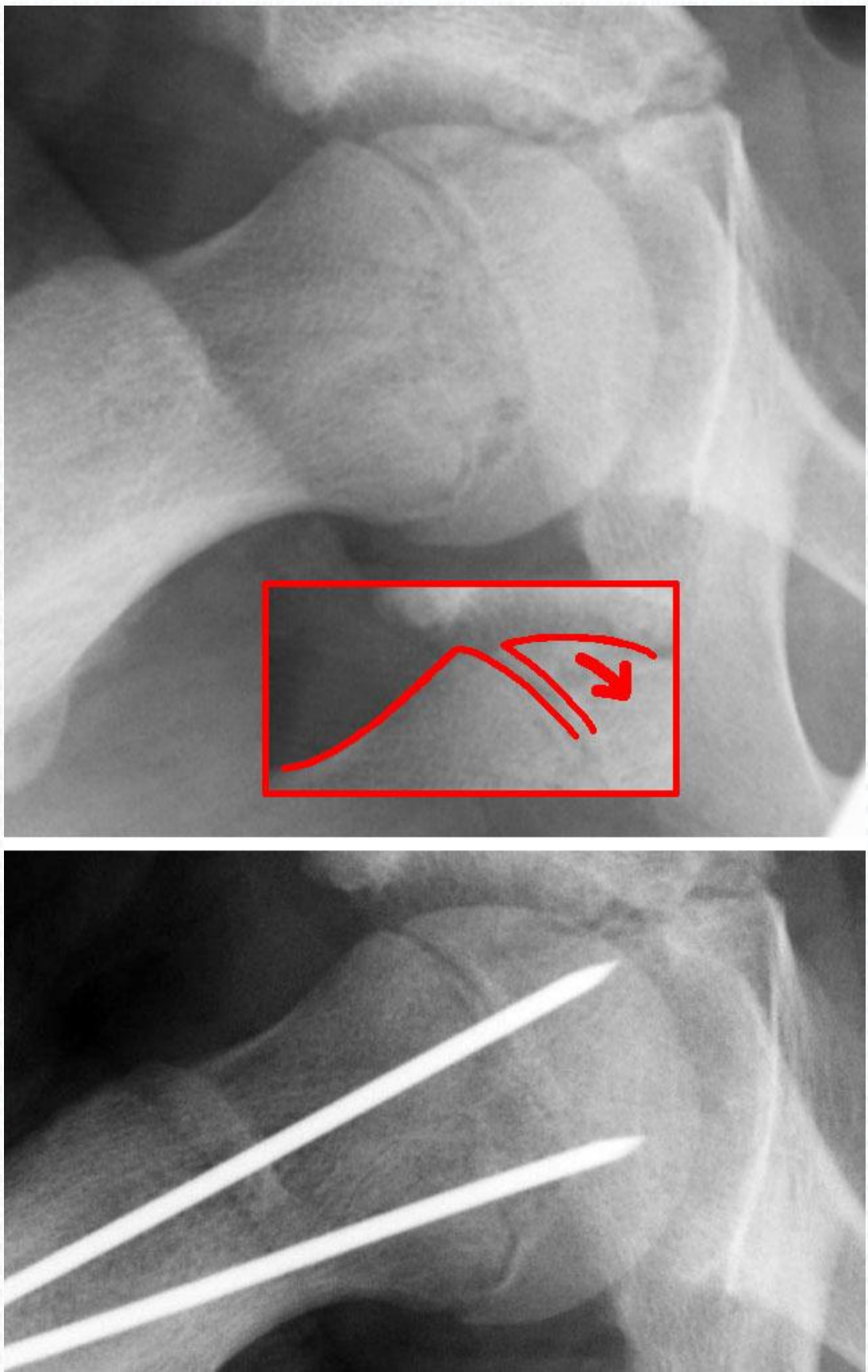


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